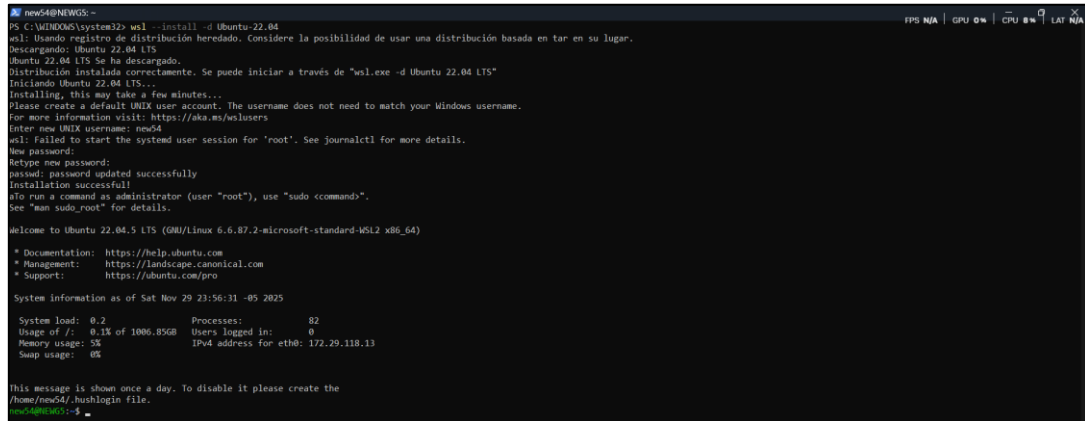


GUÍA INSTALACIÓN CASSANDRA 5 Y CONEXIÓN CON PYTHON

Paso 1: Instalación de una nueva instancia de WSL

1. Abrimos PowerShell como administrador e instalamos una instancia de *Ubuntu*

```
wsl --install -d Ubuntu-22.04
```



```
new54@NEWG5:~  
PS C:\WINDOWS\system32 wsl --install -d Ubuntu-22.04  
wsl: usando registro de distribución heredado. Considere la posibilidad de usar una distribución basada en tar en su lugar.  
Descargando: Ubuntu 22.04 LTS  
Ubuntu 22.04 LTS Se ha descargado.  
Distribución instalada correctamente. Se puede iniciar a través de "wsl.exe -d Ubuntu 22.04 LTS"  
Iniciando Ubuntu 22.04 LTS...  
Installing, this may take a few minutes...  
Please create a default UNIX user account. The username does not need to match your Windows username.  
For more information visit: https://aka.ms/wslusers  
Enter new UNIX username: new54  
wsl: failed to start the systemd user session for 'root'. See journalctl for more details.  
New password:  
Retype new password:  
passwd: password updated successfully  
Installation successful!  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:        https://ubuntu.com/pro  
  
System information as of Sat Nov 29 23:56:31 -05 2025  
  
System load: 0.2          Processes:            82  
Usage of /:  0.1% of 1006.85GB   Users logged in:     0  
Memory usage: 5%           IP address for eth0: 172.29.118.13  
Swap usage:  0%  
  
This message is shown once a day. To disable it please create the  
/home/new54/.hushlogin file.  
new54@NEWG5:~$
```

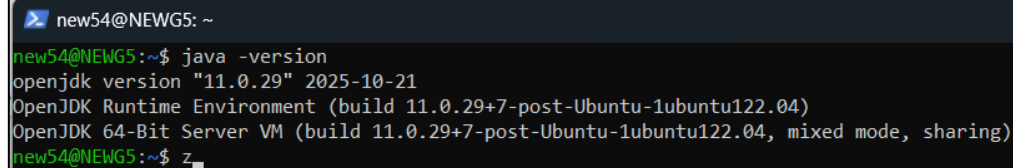
2. Actualizamos el sistema

```
sudo apt update && sudo apt upgrade -y
```

Paso 2: Instalación de Cassandra

1. Instalación de Java (OpenJDK11)

```
sudo apt install openjdk-11-jdk -y
```



```
new54@NEWG5:~  
new54@NEWG5:~$ java -version  
openjdk version "11.0.29" 2025-10-21  
OpenJDK Runtime Environment (build 11.0.29+7-post-Ubuntu-1ubuntu122.04)  
OpenJDK 64-Bit Server VM (build 11.0.29+7-post-Ubuntu-1ubuntu122.04, mixed mode, sharing)  
new54@NEWG5:~$ z_
```

2. Añadimos el repositorio de Cassandra 5

```
curl -o /etc/apt/keyrings/apache-cassandra.asc  
https://downloads.apache.org/cassandra/KEYS  
  
echo "deb [signed-by=/etc/apt/keyrings/apache-cassandra.asc]  
https://deb.debian.org/debian/cassandra 5.0x main" | sudo tee  
/etc/apt/sources.list.d/cassandra.sources.list
```

```
new54@NEWG5:/mnt/c/WINDOWS/System32$ sudo curl -o /etc/apt/keyrings/apache-cassandra.asc https://downloads.apache.org/cassandra/KEYS
% Total    % Received % Xferd  Average Speed   Time    Time     Current
           Dload  Upload   Total   Spent    Left     Speed
100 271k  100 271k    0     0  197k      0  0:00:01  0:00:01 --:--:-- 198k
new54@NEWG5:/mnt/c/WINDOWS/System32$ echo "deb [signed-by=/etc/apt/keyrings/apache-cassandra.asc] https://deb.debian.cassandra.apache.org 50x main" | sudo tee /etc/apt/sources.list.d/cassandra.sources.list
deb [signed-by=/etc/apt/keyrings/apache-cassandra.asc] https://deb.debian.cassandra.apache.org 50x main
new54@NEWG5:/mnt/c/WINDOWS/System32$ sudo apt-get update
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:3 https://apache.jfrog.io/artifactory/cassandra-deb 50x InRelease [3902 B]
Get:6 https://apache.jfrog.io/artifactory/cassandra-deb 50x/main amd64 Packages [694 B]
Fetched 305 kB in 2s (202 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  cassandra-tools
The following NEW packages will be installed:
  cassandra
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 70.5 MB of archives.
After this operation, 83.0 MB of additional disk space will be used.
Get:1 https://apache.jfrog.io/artifactory/cassandra-deb 50x/main amd64 cassandra all 5.0.6 [70.5 MB]
Fetched 70.5 MB in 10s (7037 kB/s)
Selecting previously unselected package cassandra.
(Reading database ... 44835 files and directories currently installed.)
Preparing to unpack .../cassandra_5.0.6_all.deb ...
Unpacking cassandra (5.0.6) ...
Setting up cassandra (5.0.6) ...
Adding group 'cassandra' (GID 118) ...
Done.
```

Nota: Si da error al añadir las keys usemos “sudo” antes del comando

3. Instalamos Cassandra

```
sudo apt-get update
sudo apt-get install cassandra

new54@NEWG5:/mnt/c/WINDOWS/System32$ cassandra -v
5.0.6
```

4. Iniciamos el servicio

```
sudo service cassandra start
```

5. Verificamos el servicio

```
nodetool status

new54@NEWG5:/mnt/c/WINDOWS/System32$ sudo service cassandra start
new54@NEWG5:/mnt/c/WINDOWS/System32$ nodetool status
Datacenter: datacenter1
=====
Status=Up/Down
-/ State=Normal/Leaving/Joining/Moving
-- Address      Load        Tokens       Owns (effective)  Host ID                               Rack
UN  127.0.0.1     114.71 KiB   16           100.0%            23a1716f-e277-4466-813f-f98b21e7cdc7  rack1

new54@NEWG5:/mnt/c/WINDOWS/System32$ |
```

Paso 3: Creación de una base de datos

1. Entramos en la consola de Cassandra

```
cqlsh
```

2. Creación de la base de datos

```
CREATE KEYSPACE reddit_analytics
WITH replication = {'class': 'SimpleStrategy', 'replication_factor':
1};

USE reddit_analytics;
```

```
CREATE TABLE youtube_live_comments (
    video_id text,
    fecha timestamp,
    autor text,
    comentario text,
    sentimiento float,
    PRIMARY KEY ((video_id), fecha, autor)
) WITH CLUSTERING ORDER BY (fecha DESC);
```

3. Salimos de la consola

```
exit;
```

Paso 4: Configuración de Cassandra para la conexión desde python

1. Editamos la configuración del archivo de Cassandra en WSL

```
sudo nano /etc/cassandra/cassandra.yaml
```

2. Editamos las siguientes líneas

```
rpc_address: localhost -> rpc_address: 0.0.0.0
broadcast_rpc_address: 1.2.3.4 -> broadcast_rpc_address:
<WSL_IP_ADDRESS>
native_transport_port: 9042 #(Asegurarse que este descomentado)
```

```
#
# For security reasons, you should not expose this port to the internet. Firewall it if needed.
rpc_address: 0.0.0.0
```

```
GNU nano 6.2 /etc/cassandra/cassandra.yaml
# be set to 0.0.0.0. If left blank, this will be set to the value of
# rpc_address. If rpc_address is set to 0.0.0.0, broadcast_rpc_address must
# be set.
broadcast_rpc_address: 127.0.0.1
```

```
# enable or disable keepalive on rpc/native connections
rpc_keepalive: true
```

```
# The address on which the native transport is bound is defined by rpc_address.
start_native_transport: true
# port for the CQL native transport to listen for clients on
# For security reasons, you should not expose this port to the internet. Firewall it if needed.
native_transport_port: 9042
```

Nota: Podemos usar ctrl + w para buscar en el archivo

3. Reiniciamos Cassandra

```
sudo service cassandra restart
```

Paso 5: Conexión a Cassandra desde Python (3.13.5)

1. Copiamos el siguiente código

```
from cassandra.cluster import Cluster
from cassandra.policies import RoundRobinPolicy
```

```

try:
    cluster = Cluster(
        ['127.0.0.1'],
        protocol_version=4,
        load_balancing_policy=RoundRobinPolicy()
    )
    print("Conexión al cluster creada correctamente")
except Exception as e:
    print(" Error al crear el cluster:", e)
    cluster = None

session = None
if cluster:
    try:
        session = cluster.connect('reddit_analytics')
        print("Conectado al keyspace 'reddit_analytics'")
    except Exception as e:
        print("Error al conectar al keyspace:", e)

if session:
    try:
        select_stmt = "SELECT * FROM youtube_live_comments LIMIT 40"
        rpta = session.execute(select_stmt)

        for fila in rpta:
            print(fila)
    except Exception as e:
        print("Error al ejecutar la consulta:", e)

if cluster:
    cluster.shutdown()

```

Nota: Debemos cambiar el nombre del keyspace de acuerdo con el nombre que hayamos asignado a nuestra tabla en el paso 3, asimismo la consulta debe ser de acuerdo con lo que necesitamos y a los nombres de las tablas

Nota: 2 Si no insertamos nada no se imprimirá nada en la consola

Paso 6 (Opcional): Insercción de datos a una tabla de Cassandra mediante Python con el API de youtube

1. Creamos nuestra API

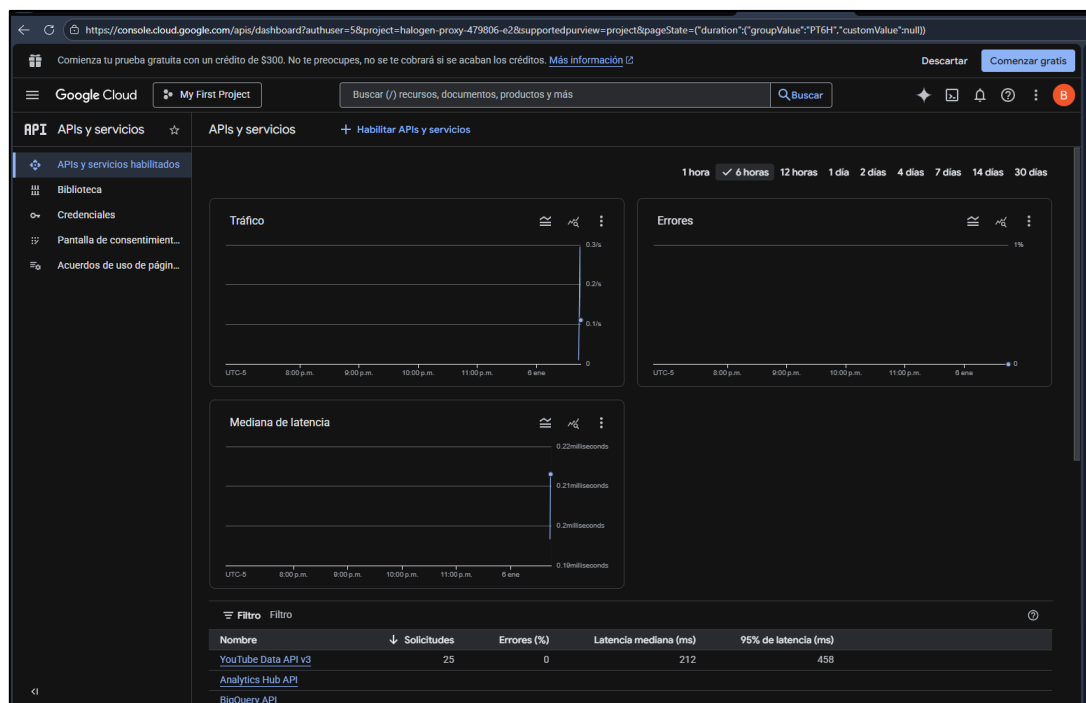
The screenshot shows the Google Cloud console interface for the YouTube Data API v3. The page is titled "YouTube Data API v3" and includes a description: "The YouTube Data API v3 is an API that provides access to YouTube data, such as videos, playlists, and channels...". It also features a "Descripción general" section with more details about the API and a "Detalles adicionales" section with technical specifications. The API is shown as "API habilitada" (API enabled).

Descripción general

The YouTube Data API v3 is an API that provides access to YouTube data, such as videos, playlists, and channels.

Detalles adicionales

Tipo: SaaS & APIs
Última actualización del producto: 21/7/22
Categoría: YouTube
Nombre del servicio: youtube.googleapis.com



2. Inserción de datos desde Python

```
from dotenv import load_dotenv
import os
import time
from dateutil import parser
from textblob import TextBlob
```

```

from googleapiclient.discovery import build
from cassandra.cluster import Cluster
from cassandra.policies import RoundRobinPolicy

load_dotenv('Credenciales.env')
YOUTUBE_API_KEY = os.getenv('API_KEY')

ALBUM_VIDEOS = {
    'Never Gonna Give You Up': 'dQw4w9WgXcQ'
}

MAX_COMMENTS_PER_VIDEO = 1000000

cluster = Cluster(['127.0.0.1'], protocol_version=4,
load_balancing_policy=RoundRobinPolicy())
session = cluster.connect('reddit_analytics')

insert_stmt = session.prepare("""
    INSERT INTO youtube_live_comments (video_id, fecha, autor, comentario,
sentimiento)
    VALUES (?, ?, ?, ?, ?)
""")

select_stmt = session.prepare("""
    SELECT fecha, autor, comentario, sentimiento
    FROM youtube_live_comments
    WHERE video_id = ?
    ORDER BY fecha DESC
    LIMIT 5
""")

youtube = build('youtube', 'v3', developerKey=YOUTUBE_API_KEY)

for song_name, video_id in ALBUM_VIDEOS.items():
    print(f"\n--- Procesando: {song_name} ({video_id}) ---")

    total_procesados = 0
    token = None

    while total_procesados < MAX_COMMENTS_PER_VIDEO:
        try:
            request = youtube.commentThreads().list(
                part="snippet",
                videoId=video_id,
                maxResults=100,

```

```

        pageToken=token,
        textFormat="plainText",
        order="relevance"
    )
    response = request.execute()

    batch_count = 0
    for item in response['items']:
        comment =
item['snippet']['topLevelComment']['snippet']
        texto = comment['textDisplay']
        score = TextBlob(texto).sentiment.polarity

        try:
            session.execute(insert_stmt, [
                video_id,
                parser.parse(comment['publishedAt']),
                comment['authorDisplayName'],
                texto[:100],
                float(score)
            ])
            batch_count += 1
        except Exception:
            pass

    total_procesados += batch_count
    print(f"    -> Guardados {batch_count} comentarios
(Total: {total_procesados})")

    if 'nextPageToken' in response:
        token = response['nextPageToken']
    else:
        break

    time.sleep(0.5)

except Exception as e:
    print(f"    [X] Error: {e}")
    break

print(f"\n--- Últimos 5 comentarios registrados para {song_name}
---")
try:
    rows = session.execute(select_stmt, [video_id])
    for row in rows:

```

```
print(f"[{row.fecha}] {row.autor}: {row.comentario}  
(Sent: {row.sentimiento})")  
except Exception as e:  
print(f" [!] Error al consultar: {e}")
```

```
(youtubeAnalytics) PS D:\UNC\Ciclo 2025-II (VI)\Tópicos especiales en Inteligencia de Negocios\Unidad II\Semana 8> & "D:/UNC/Ciclo 2025-II (VI)/Tópicos especiales en Inteligencia de Negocios/Unidad II/Semana 8/youtubeAnalytics/Scripts/python.exe" "d:/UNC/Ciclo 2025-II (VI)/Tópicos especiales en Inteligencia de Negocios/Unidad II/Semana 8/InsercionDeDatosCassandra.py"  
  
--- Procesando: Never Gonna Give You Up (dQw4w9WgXcQ) ---  
-> Guardados 100 comentarios (Total: 100)  
-> Guardados 100 comentarios (Total: 200)  
-> Guardados 100 comentarios (Total: 300)  
-> Guardados 100 comentarios (Total: 400)  
-> Guardados 100 comentarios (Total: 500)  
-> Guardados 100 comentarios (Total: 600)  
-> Guardados 100 comentarios (Total: 700)  
-> Guardados 100 comentarios (Total: 800)  
-> Guardados 100 comentarios (Total: 900)  
-> Guardados 100 comentarios (Total: 1000)  
-> Guardados 100 comentarios (Total: 1100)  
-> Guardados 47 comentarios (Total: 1147)  
  
--- Últimos 5 comentarios registrados para Never Gonna Give You Up ---  
[2026-01-06 06:42:33] @fentwoent: who got brought here by Ella Langley ♡(Sent: 0.0)  
[2026-01-06 06:34:05] @michelinadicco8253: January 2026♥(Sent: 0.0)  
[2026-01-06 06:22:12] @Ng0Linh-hdu: 🍷🍷 (Sent: 0.0)  
[2026-01-06 06:15:47] @Imaztly777: My friend trolled me and I ate my first rick roll in 2026 🍷🍷 (Sent: 0.25)  
[2026-01-06 06:09:32] @popofsergel8766: in 2026, shame on me... (Sent: 0.0)  
(youtubeAnalytics) PS D:\UNC\Ciclo 2025-II (VI)\Tópicos especiales en Inteligencia de Negocios\Unidad II\Semana 8> []
```